SPIRE Universal Ingest Service

Companion to the SPIRE Commander Brief

BLUF. The Universal Ingest Service is the data intake layer underneath SPIRE. It accepts every common file type from every common source, maps the four core Marine logistics systems out of the box, and uses an on-board AI model running entirely offline to handle unfamiliar files. Every record landing in the canonical store gets a tamper-evident audit entry. UIS is built to be extracted from SPIRE and used on its own. No commercial or federal product surveyed combines all four of those properties.

What it does

- **Reads anything tabular.** CSV, spreadsheets, JSON, XML, fixed-width legacy exports, even EDI X12 supply transactions.
- **From any common source.** Drag a file in, pull from a secure share, watch an inbox for attachments, poll a web service, poll a database, or subscribe to a streaming feed. Every source retries on its own when networks blink and isolates malformed input so it can't poison the rest of the data.
- **Marine logistics is pre-mapped.** GCSS-MC asset roster, utilisation, and service requests; DRRS-MC unit readiness ratings. Adding a fifth source is a one-day engineering task.
- **On-board AI for unfamiliar files.** When a unit drops a file that doesn't match a known system, an AI model running entirely on the same 2U box reads the headers and proposes how each column should map. Sample data is scrubbed before the model sees it. The operator confirms once; the mapping saves so the unit doesn't redo the work next time.
- **Audit on every write.** Every record committed to the canonical store is cryptographically chained. An inspector can verify the chain offline against a public key. Feeds existing security monitoring tools.

What is distinctive

A 2026 survey of commercial data-integration platforms (Informatica, Talend, Airbyte, Fivetran), federal data systems (Palantir Foundry, ADVANA, Maven Smart System, the CDAO Edge Data Mesh), and the leading open-source stack (Apache NiFi) found three properties that SPIRE has and they don't:

- **Marine systems pre-mapped on day one.** GCSS-MC and DRRS-MC arrive ready to ingest. No commercial or federal product surveyed ships with this pre-built; everyone else expects you to build the mapping yourself, typically as a contractor engagement.
- **Tamper-evident at the moment of write.** Every record committed to the canonical store is cryptographically chained at the moment of ingest. Other ingest stacks have audit logs; SPIRE has a chain that an inspector can verify offline with no access to the running system.
- **Built for the operator, not the database administrator.** A unit S-4 can ingest an unfamiliar file without calling a contractor. The AI lives on the same 2U box; sample data is scrubbed before the model sees it; the resulting mapping saves so the same unit doesn't redo the work.

Honest framing

Reading lots of file types from lots of sources is not new. AI-assisted column mapping is not new. Running offline is not new. UIS is not the only universal ingest service in DoD — Foundry, NiFi, and the CDAO Edge Data Mesh all read data. UIS does not replace them. It is the unit-side, last-mile layer that turns whatever the operator gets handed into canonical Marine logistics rows. Higher-tier systems consume those rows downstream.

Extractable from SPIRE

UIS was built from the start to be liftable — the modularisation work was a deliberate engineering goal, not an afterthought. Estimated effort to ship UIS as a standalone product, separate from SPIRE: **one engineer-day**. The Marine logistics adapters, the offline AI mapper, and the tamper-evident audit chain travel with it.

Maturity

TRL 4 as part of SPIRE; validated at Modern Day Marine 2026 against a synthetic GCSS-MC dataset shaped to match the live export. The web, database, and streaming source types are scaffolded but need production hardening. No ATO yet.

Path forward

- **A sanitised real GCSS-MC export** from a sponsoring MEF G-4 or MLR S-4. The synthetic dataset proves shape; only a real pull proves edge cases.
- **An MCSC POC** to scope UIS as either a SPIRE-internal component or a separately licensable Marine-Corps logistics ingest library — architecture supports either.

POC: Jesse Morgan · Thornveil · jesse@thornveil.ai
Code: github.com/jeranaias/spire/tree/master/backend/uis (public)

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